

PROJECT SUMMARY REPORT

VAGDEVI MALINENI

Project Title

Raritone – AI-Powered Virtual Try-On Platform

Introduction

Raritone is an AI-powered virtual try-on platform designed to enhance the online fashion shopping experience. The platform allows users to create avatars, upload body measurements, browse products, and visualize how garments would look on them before making a purchase. The system combines modern backend technologies, cloud-based media management, real-time communication, and AI-driven recommendations to deliver a seamless and personalized user experience.

Project Objective

The primary objective of the project is to develop a scalable backend system that supports virtual try-on functionality, user management, product management, media processing, and AI-powered size recommendations.

The project aims to:

- Improve online shopping confidence through virtual garment visualization.
- Provide personalized size recommendations based on body measurements.
- Optimize image storage and delivery using cloud technologies.
- Support real-time communication and future AI integrations.
- Build a secure, scalable, and maintainable backend architecture.

Features Implemented

User Management

- User Registration
- User Login
- JWT Authentication
- Role-Based Authorization
- Protected Routes

Product Management

- Product CRUD Operations
- Product Image Upload
- Product Categorization
- Inventory Management

Avatar Management

- Avatar Upload and Storage
- Avatar Retrieval APIs
- User Avatar Mapping

Measurement Management

- Body Measurement Storage
- Measurement Retrieval APIs
- User-Specific Measurement Tracking

Virtual Try-On Preparation

- Try-On Database Design
- Product Mapping Design
- AI Workflow Planning

Media Management

- Multer File Uploads
- Sharp Image Compression
- Cloudinary Integration
- CDN-Based Image Delivery

Security Features

- JWT Authentication
- Joi Validation
- Helmet Security Headers
- Morgan Request Logging
- Rate Limiting
- Centralized Error Handling

Real-Time Communication

- Socket.IO Integration Planning
- Real-Time Notifications
- Live Processing Updates

Technologies Used

Backend Technologies

- Node.js
- Express.js

Database

- MongoDB Atlas
- Mongoose ODM

Authentication & Security

- JWT (JSON Web Tokens)
- Joi Validation
- Helmet
- Express Rate Limit
- Morgan

Media Processing

- Multer
- Sharp
- Cloudinary

Real-Time Communication

- Socket.IO

Deployment & Cloud

- Render
- MongoDB Atlas
- Cloudinary CDN

Monitoring & Analytics

- Sentry
- Grafana
- Firebase Analytics

Development Tools

- Postman
- Git
- GitHub
- VS Code

Backend Modules

Authentication Module

Responsible for user registration, login, token generation, and secure route access.

Functions

- Register User
- Login User
- Verify JWT
- Role Authorization

Product Module

Manages all product-related operations.

Functions

- Add Product
- Update Product
- Delete Product
- View Products

Avatar Module

Handles avatar uploads and avatar-related operations.

Functions

- Upload Avatar
- Store Avatar URL
- Retrieve Avatar Data

Measurement Module

Stores user body measurements for future AI processing.

Functions

- Add Measurements
- Update Measurements
- Retrieve Measurements

Media Management Module

Processes and optimizes uploaded images.

Functions

- Upload Files
- Compress Images
- Store Images in Cloudinary
- Deliver Optimized Images

Security Module

Provides protection against unauthorized access and malicious requests.

Functions

- JWT Authentication
- Request Validation
- Rate Limiting
- Error Handling

Future AI Module

Planned AI system for virtual try-on and size prediction.

Planned Functions

- Body Analysis
- Garment Overlay
- Size Recommendation
- Personalized Suggestions

Results

The project successfully established a strong backend foundation for an AI-powered virtual try-on platform.

Achievements

- ✓ Secure user authentication and authorization
- ✓ Product management APIs implemented
- ✓ Avatar and measurement data management completed
- ✓ Media upload and optimization workflow designed
- ✓ Cloud-based storage integration completed
- ✓ Security middleware implementation verified

- ✓ Scalable database architecture designed
- ✓ AI integration workflow planned
- ✓ Deployment and monitoring strategies researched
- ✓ Future scaling architecture prepared

Future Enhancements

The following improvements are planned for future development:

AI Features

- AI-Based Virtual Try-On Engine
- Body Segmentation Models
- Pose Estimation
- Garment Overlay Processing

Recommendation System

- Personalized Product Recommendations
- Smart Size Prediction
- Fit Score Calculation

Performance Improvements

- Redis Caching
- Advanced CDN Optimization
- Database Query Optimization

Scalability Enhancements

- Microservices Architecture
- Load Balancing
- Auto Scaling Infrastructure
- Multi-Region Deployment

User Experience

- Real-Time Try-On Progress Updates

- Advanced Search & Filters
- Wishlist Management
- Order Tracking

Monitoring

- Sentry Error Tracking
- Grafana Dashboards
- Firebase Analytics Integration

Conclusion

The Raritone project successfully developed a secure, scalable, and cloud-ready backend architecture for a virtual try-on platform. The implementation includes user management, product management, avatar handling, measurement tracking, media optimization, and security mechanisms. The architecture is designed to support future AI-powered virtual try-on capabilities, personalized recommendations, and large-scale deployment.

With the integration of modern technologies such as Node.js, MongoDB Atlas, Cloudinary, Socket.IO, and planned AI services, the platform is well-positioned for future growth and production deployment. The project provides a strong foundation for delivering an innovative and personalized online shopping experience.